Setup 06 : MC da 10^6 sample $\Rightarrow A_{06}^{down} = 0.3$ e $A_{06}^{up} = 0.3$

Setup 04 : MC da 10^6 sample $\Rightarrow A_{06}^{down} = 0.66$ e $A_{06}^{up} = 0.66$

Setup 08 : MC da 10^6 sample $\Rightarrow A_{06}^{down} = 0.70$ e $A_{06}^{up} = 0.2\bar{5}$

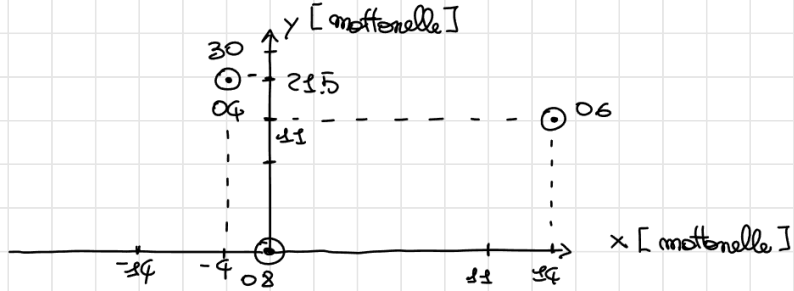
$$\Omega_{08} = 4\,\mathrm{arctg}\left(\frac{\Delta x \Delta y}{2 d_{13}\sqrt{4 d_{13}^2 + \Delta x^2 + \Delta y^2}}\right) = 4.8\ \mathrm{sr}$$

$$f = \frac{1218}{2 \cdot 3.8\sqrt{58^2 + 21^2 + 4 \cdot 3.8^2}} = \frac{1218}{2 \cdot 3.8 \cdot 62.15} = 2.58$$

$$\Omega_{06} = \frac{800\ \mathrm{cm}^2}{60.5^2\ \mathrm{cm}^2} = 0.55\ \mathrm{str}$$

$$\Omega = 4\,\mathrm{arctg}(f) = 4\,\mathrm{arctg}(2.58) = 4 \cdot 1.2 = 4.8$$

$$\Omega_{04} = \frac{2000\ \mathrm{cm}^2}{41^2\ \mathrm{cm}^2} = 1.180\ \mathrm{str}$$



$mt = 23.8 \pm 0.1\ cm$

$\sigma_{mt} = 0.25\ mt$

$08 = (0,0)$

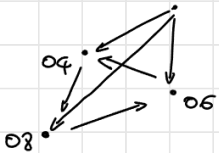
$06 = (14, 11)$

$04 = (-4, 21.5)$

$d_{08-06} = 17.8\ mt \ \& \ 5.3044\ m$

$d_{08-04} = 21.86\ mt \ \& \ 6.5170\ m$

$d_{06-04} = 20.83\ mt \ \& \ 6.2100\ m$



$z_{08} = 1.85\ m$ $\qquad \vec{x}_8 = (0, 0, 1.85)$

$z_{06} = 1.76\ m$ $\qquad \vec{x}_6 = (4.172, 3.278, 1.76)$

$z_{04} = 2.685\ m$ $\qquad \vec{x}_4 = (-1.132, 6.407, 2.685)$

$\vec{v} = \vec{x}_6 - \vec{x}_8 = (4.172, 3.278, -0.19)$

$\vec{w} = \vec{x}_4 - \vec{x}_8 = (-1.132, 6.407, -0.735)$

$$\vec{v} \times \vec{w} = \det \begin{pmatrix} \hat{x} & \hat{y} & \hat{z} \\ 4.172 & 3.278 & -0.19 \\ -1.132 & 6.407 & -0.735 \end{pmatrix} \qquad \vec{m} = \begin{pmatrix} -1.132 \\ 3.283 \\ 30.637 \end{pmatrix}$$

$$\hat{m} = \frac{\vec{m}}{30.83} = \begin{pmatrix} -0.039 \\ 0.107 \\ 0.994 \end{pmatrix}$$

$\cos\theta = 0.994 \Rightarrow \theta = 0.11\ rad$

$\sin\varphi = \dfrac{0.107}{\sin\theta} = 0.975$

$\hookrightarrow \varphi = 0.83\ rad$

I° discriminatore	II° discriminatore	I° U.L.P.	II° U.L.P.	I° NIM-TTL
PMT02 -08	PMT01 - 08	T08	T04	T08
PMT02 -08	PMT03 - 08	T06	08 & 06 & 04	D_{12} 08
PMT02 -08	PMT01 - 06			D_{23} 08
PMT01 - 04	PMT03 - 06			D_{31} 08
	T06 (doppio)			
(INFN Pisa)	T04 (doppio)			T06
	T08 (doppio)			D_{12} 06
	PMT03-04			D_{23} 06
				D_{31} 06
	(LRS)			

DEO NANO

0	T08	4	T06	8	T04
1	D_{12} 08	5	D_{12} 06	9	D_{12} 04
2	D_{23} 08	6	D_{23} 06	10	D_{23} 04
3	D_{31} 08	7	D_{31} 06	11	D_{31} 04

II° NIM-TTL

T04
D_{12} 04
D_{23} 04
D_{31} 04